UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2026-2027

LECTURE TITLE

Imaging of Pleural Diseases

DEPARTMENT OF SURGERY / RADIOLOGY SECTION

Stage 4 • Academic Year 2026–2027

Lecturer Dr. Maryam Issa

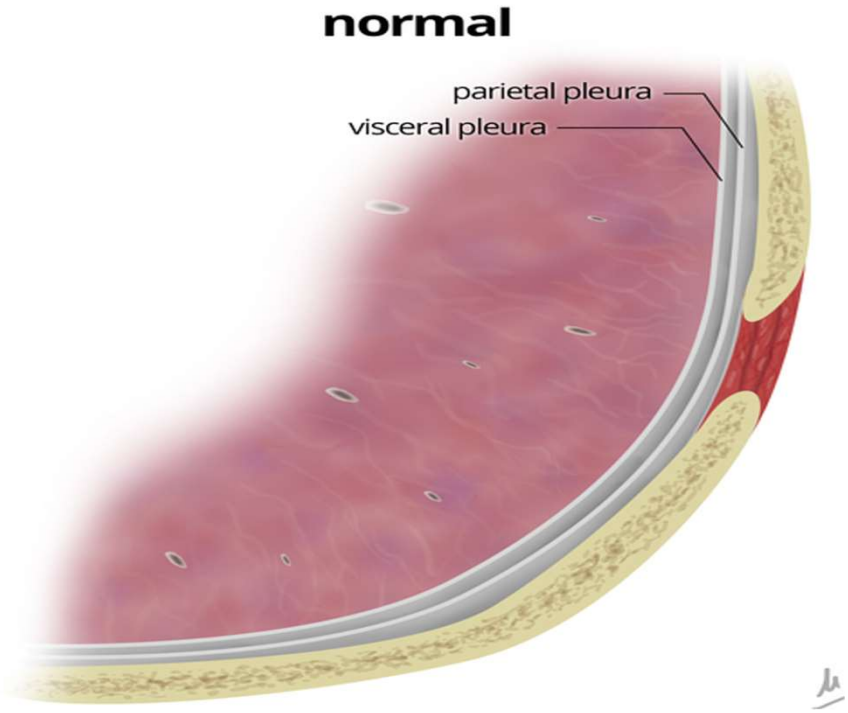
College of Medicine – University of Baghdad

Learning Objectives

- Recognize normal pleural fissures and the fluid volume thresholds required to see effusions on PA, lateral, and supine chest X-rays.
- Differentiate a **loculated empyema** from a **lung abscess** using contrast CT criteria.
- Identify acute pneumothorax on upright/supine CXR (*Deep Sulcus Sign*).
- Recognize radiological signs of **tension pneumothorax** requiring immediate decompression.

Essential Anatomy

- Is a serous membrane.
- Divided in to : visceral and parietal layers.
- Pleural space is a potential space between the two layers.



Pneumothorax

- Refers to the presence of air withing the pleural space.
- The majority of pneumothoraces occur in young people with no recognizable lung disease, small blebs or bullae at the periphery of their lungs which bursts.
- Other causes includes : emphysema , trauma, pneumocystis infection ,certain forms of interstitial pulmonary disease and rarely metastasis.

Radiological Findings

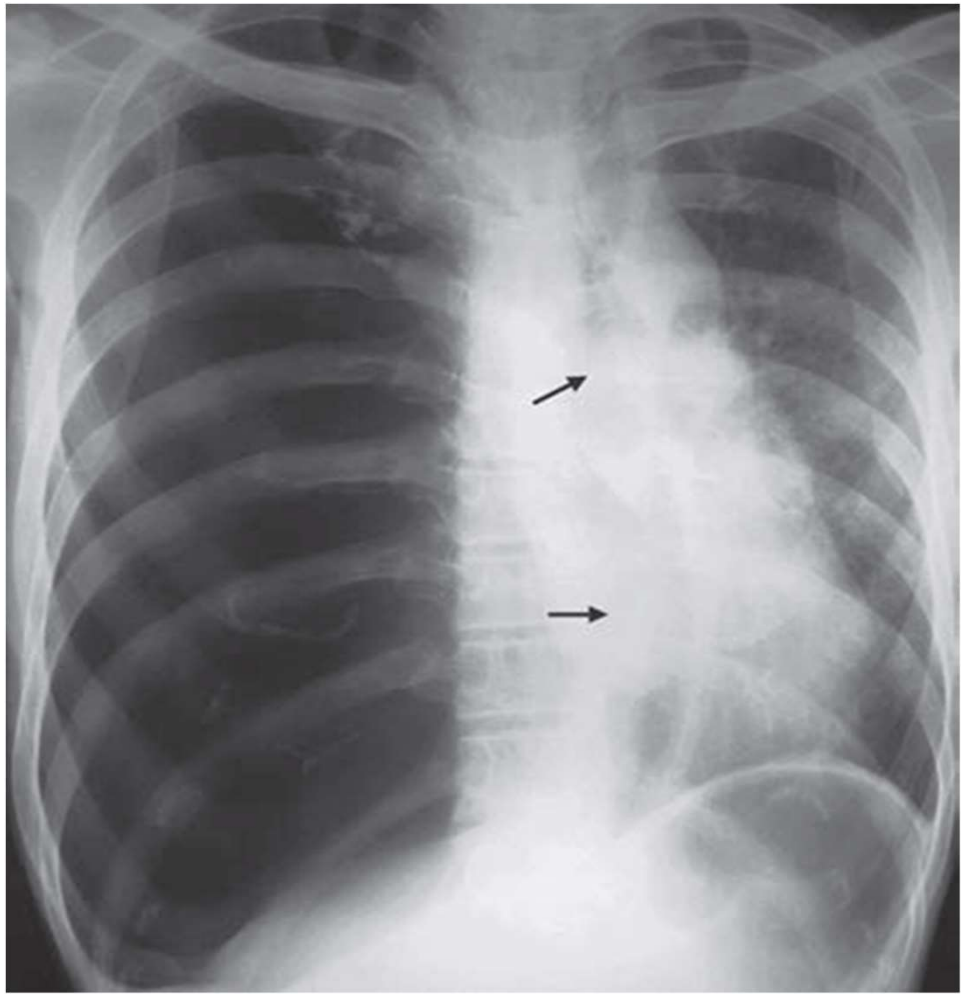
- A line of pleura due to the lung edge being separated from the chest wall, mediastinum or diaphragm by air.
- Absence of vessel shadows outside this line.
- In tension pneumothorax there is mediastinal shift and flattening of the hemidiaphragm.



Tension pneumothorax

Contralateral mediastinal shift

Flattening of the hemidiaphragm

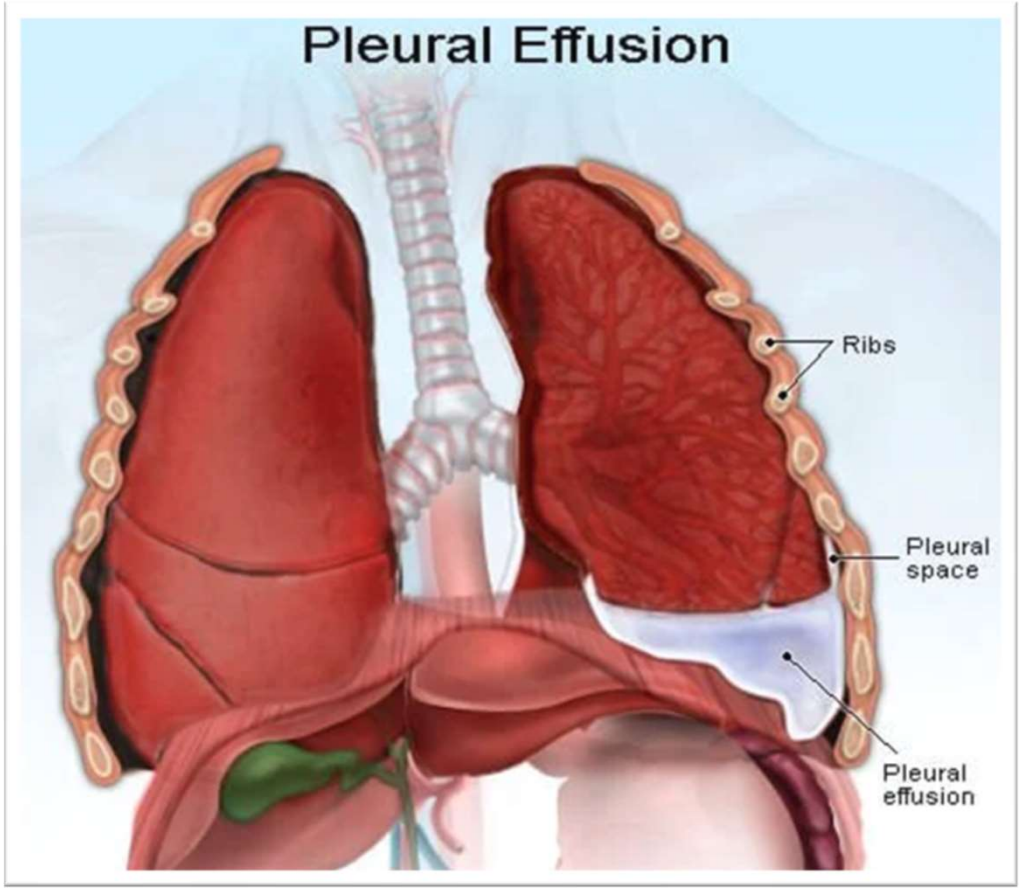


Pleural Effusion

Accumulation of the fluid in the pleural space.

Types :

1. Free effusion: which assumes the shape of the dependent thoracic cage and the shape of the underlying lung.
2. Loculated effusion: due to adhesion; particularly in empyema.

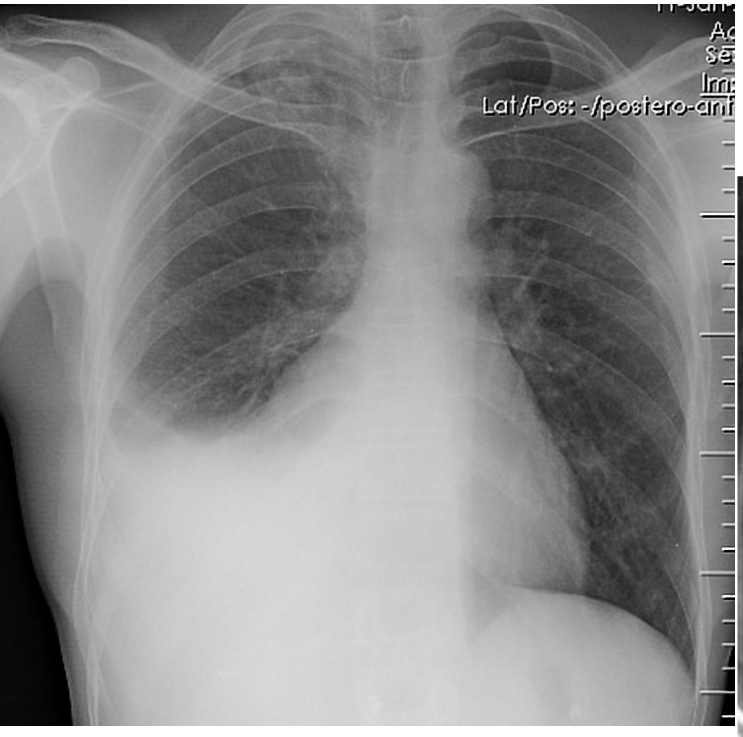


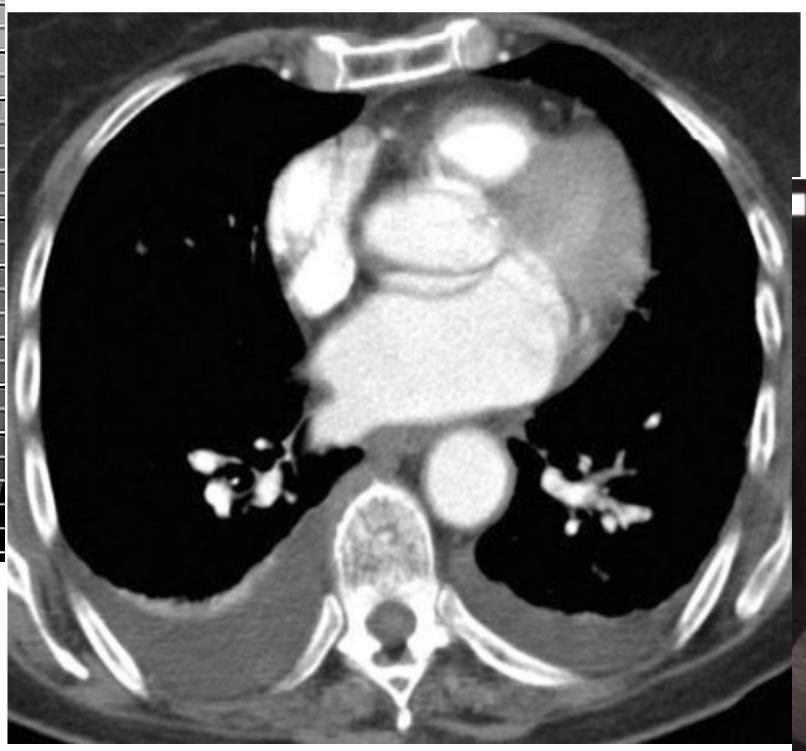
What are the common causes of pleural effusions ?

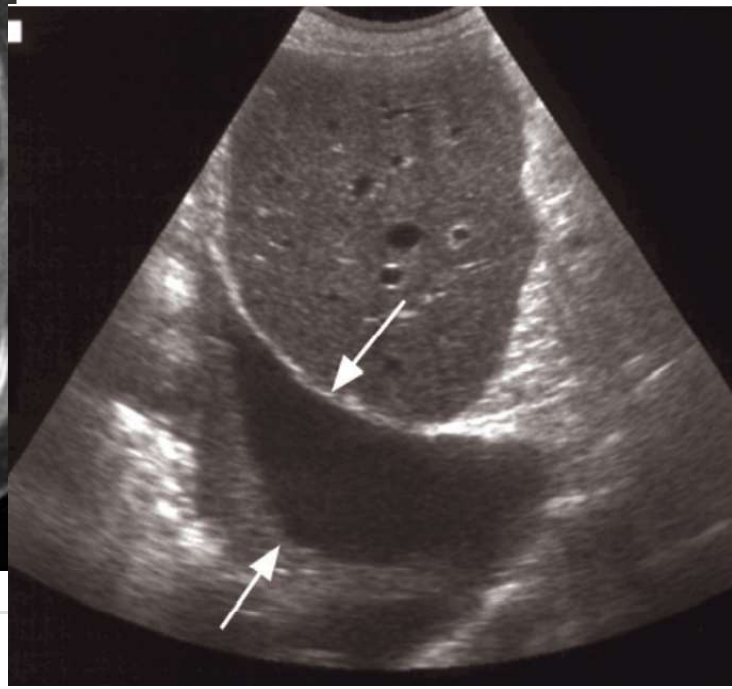
Radiographic Features

- Homogenous opacity causing obliteration of costophrenic angle.
- Usually, the fluid surrounds the lung and is higher laterally than medially (meniscus sign).
- Large effusion causing contralateral mediastinal shift.
- Loculated effusion simulating pleural mass on CXR

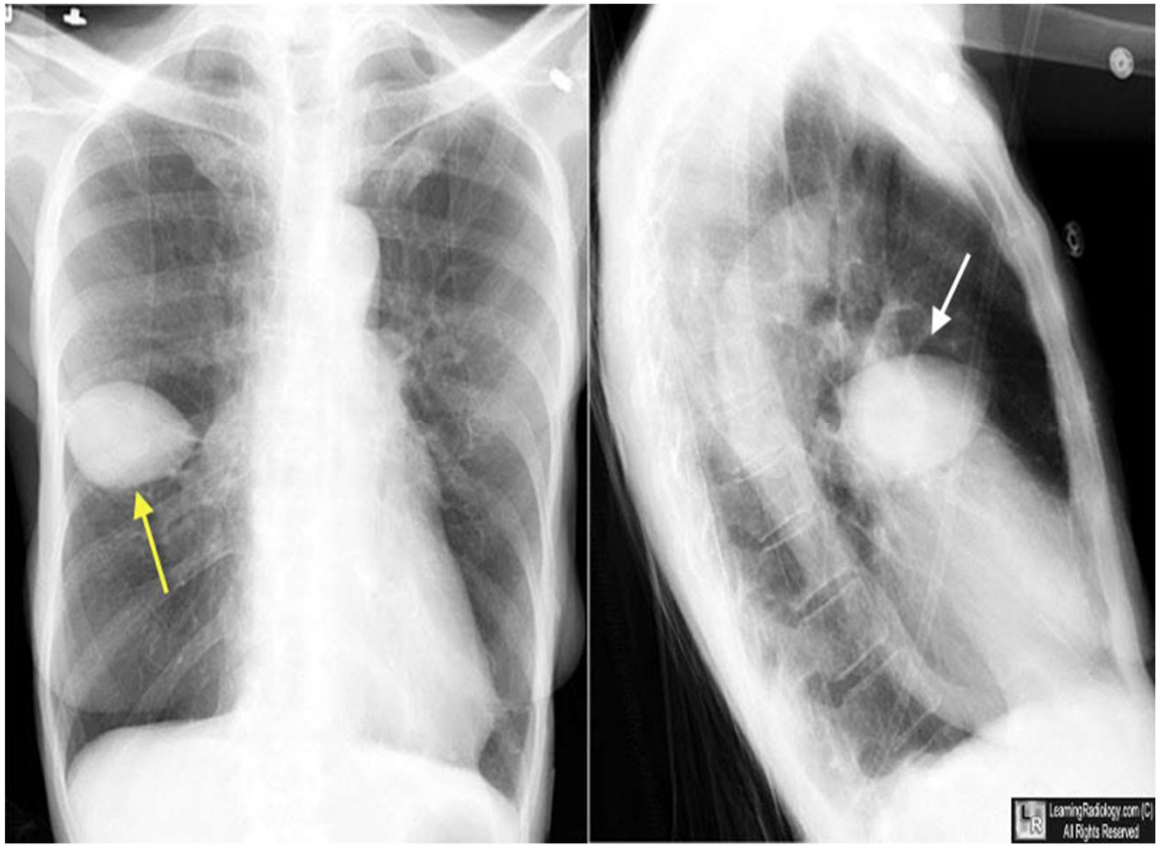
Pleural effusion

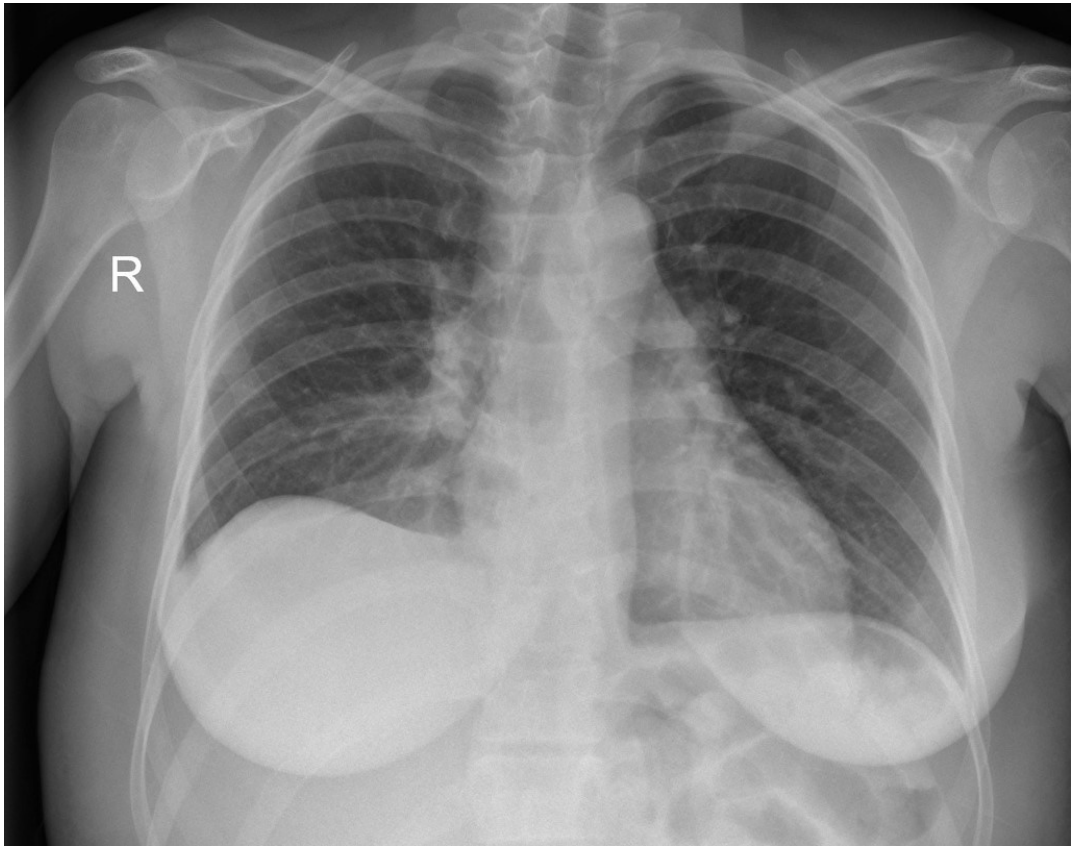






- Encysted fluid in the horizontal fissure biconvex shape which may disappear on subsequent imaging called "vanishing tumor".





Sub-pulmonary effusion simulating elevated diaphragm on CXR

Pleural empyema

- Refers to an infected purulent and often loculated pleural effusion.
- Causes large unilateral pleural collection.

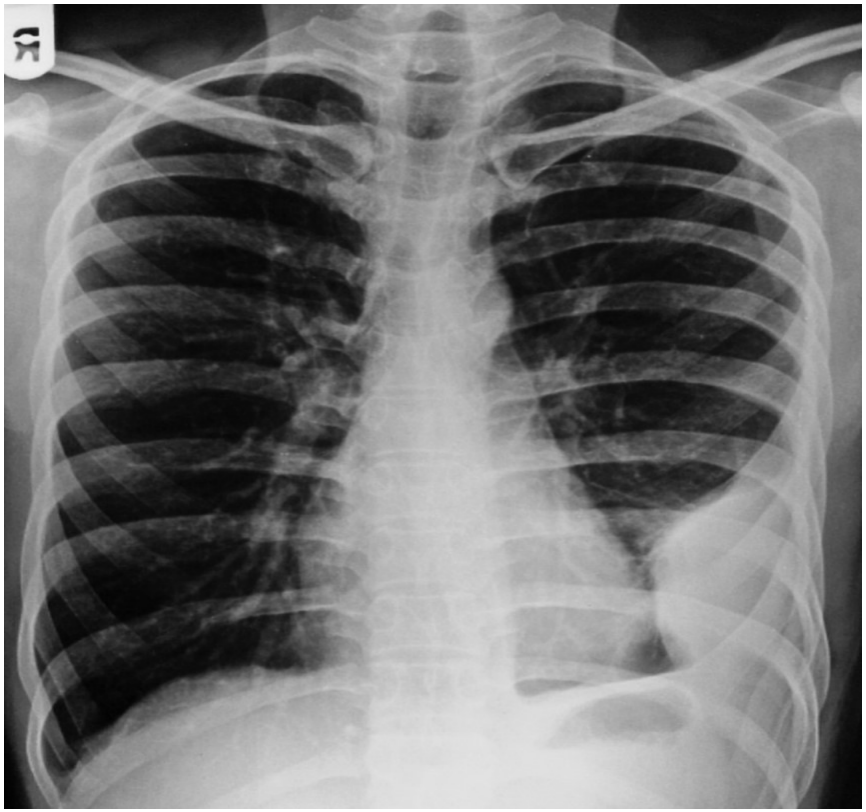
Radiological features

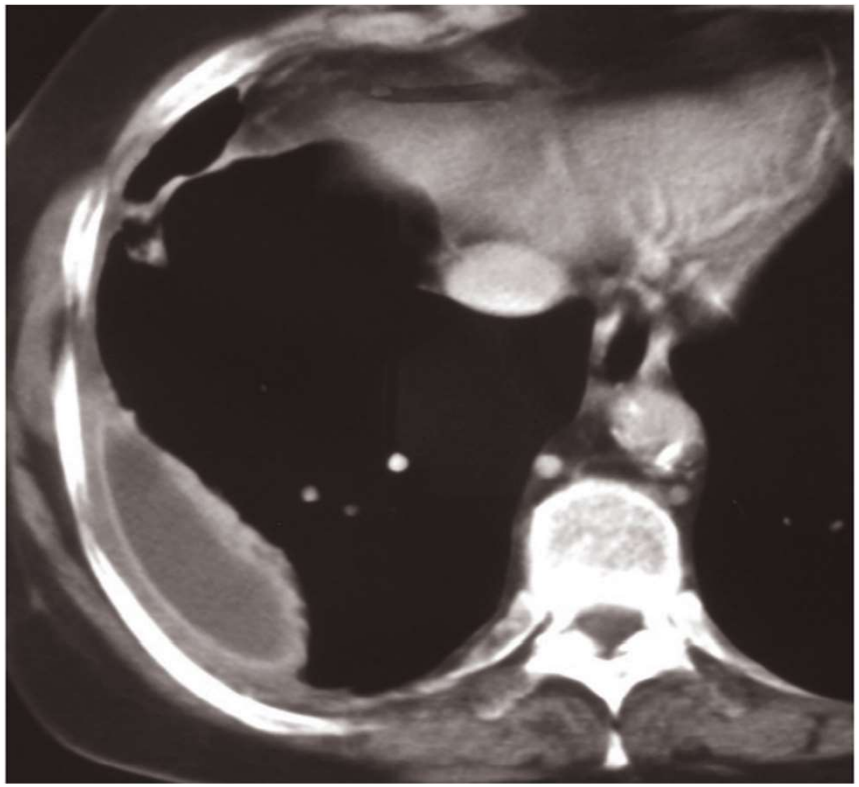
CXR:

- Lenticular shape collection simulating pleural mass.
- Forming obtuse angle with the chest wall.

CT scan :

- Lenticular shape fluid collection.
- Thick enhanced wall.
- Split pleura sign (the most sensitive and specific sign differentiating empyema from peripheral lung abscess).

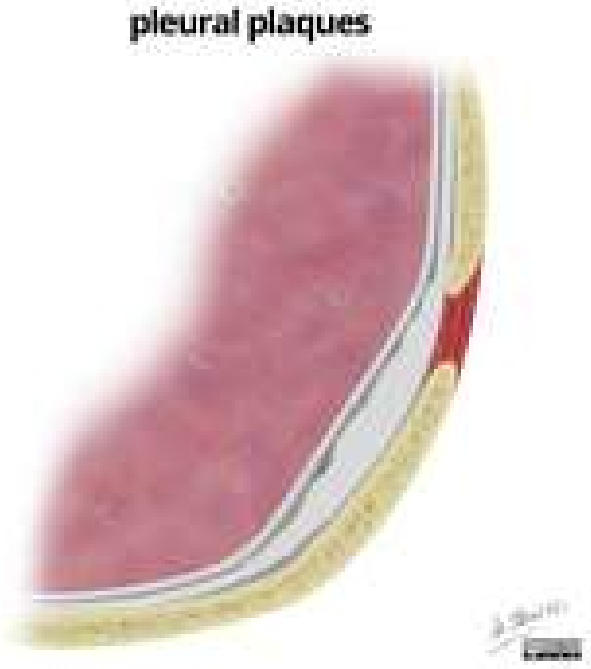


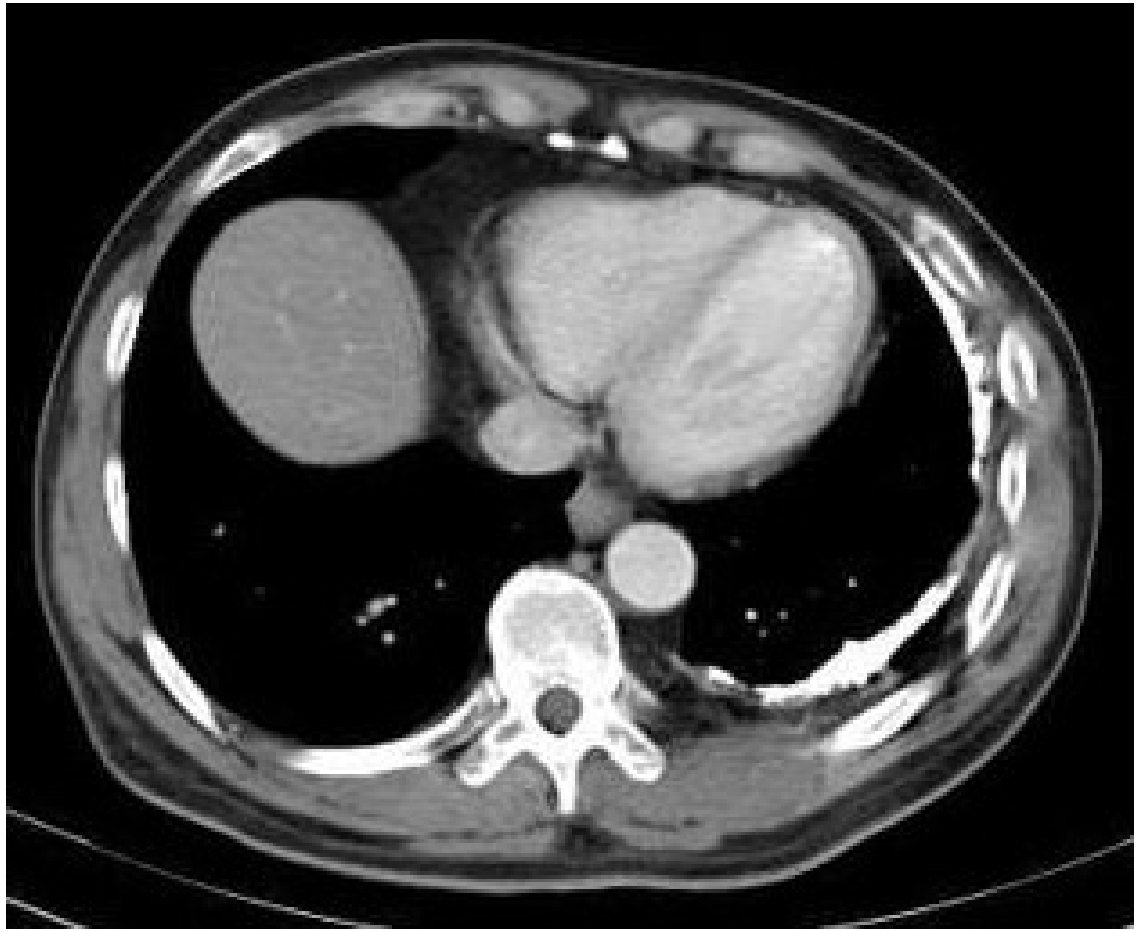


Pleural empyema

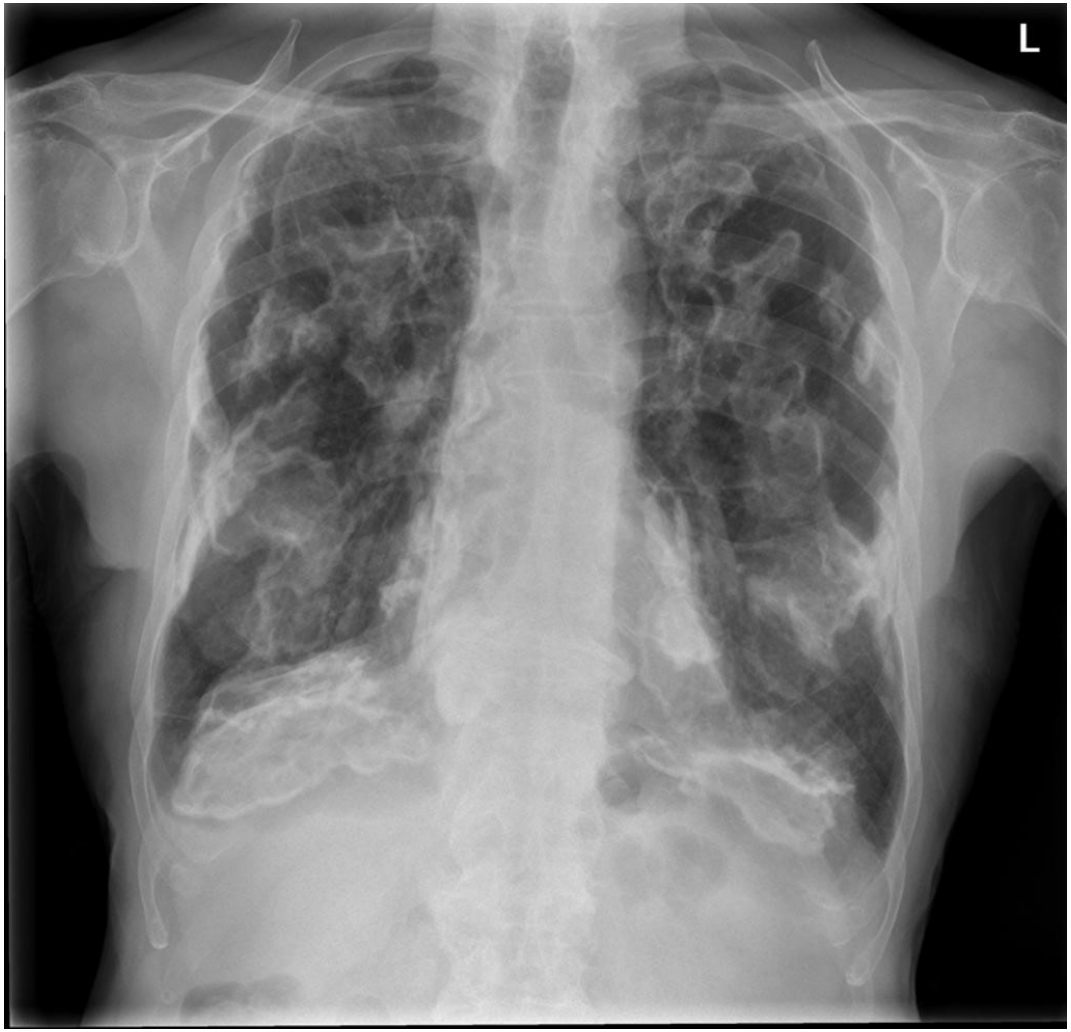
Benign Pleural Thickening

1. Following recurrent inflammation.
2. Following pleural empyema.
3. Delayed complication of hemothorax.
4. Asbestos related pleural disease.





Thickening and calcification of the pleura due to previous empyema



Diffuse pleural plaques secondary to asbestos exposure

Pleural Neoplasms

1.Benign neoplasms:

Solitary fibrous tumor.

2.Malignant neoplasms:

Mesothelioma.

Metastasis.

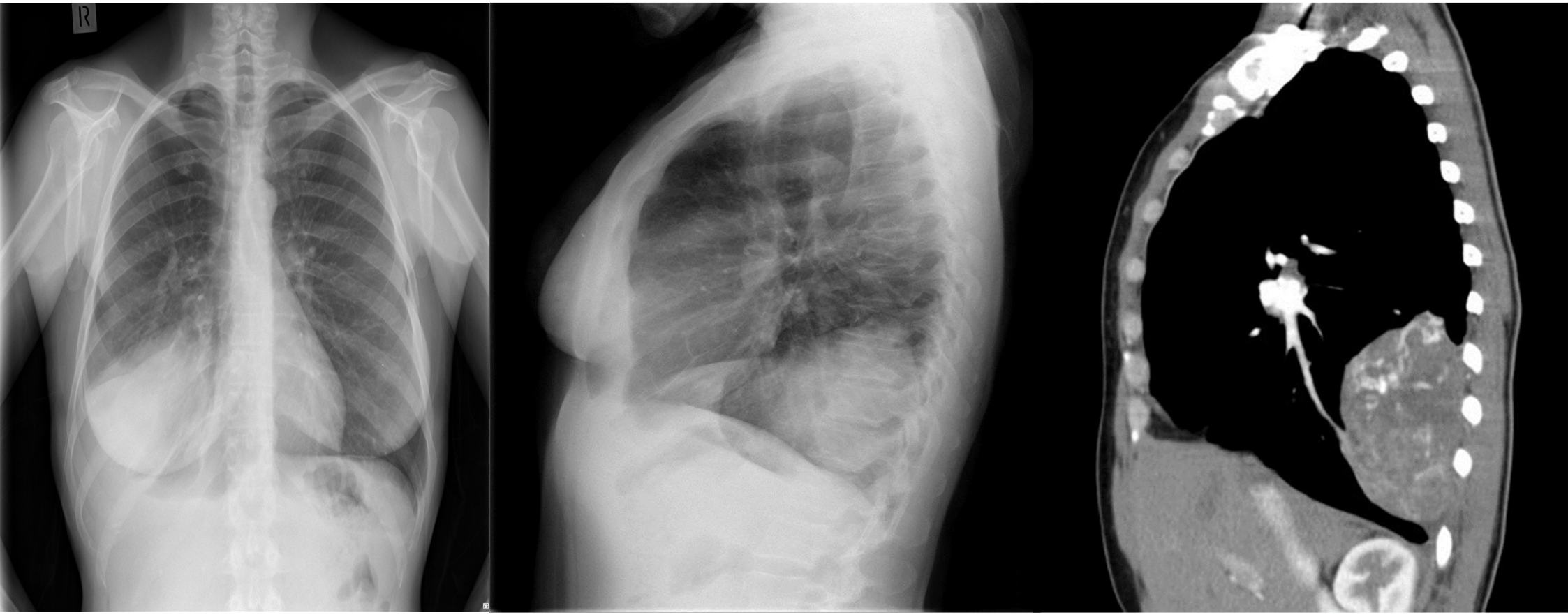
Benign neoplasms

Solitary fibrous tumors of the pleura are rare benign that account for <5% of all tumors involving the pleura.

Radiological features :

- Pleural based mass (obtuse angle with the chest wall).
- Soft tissue attenuation.
- Enahncing post contrast.
- Incomplete border sign may be present.

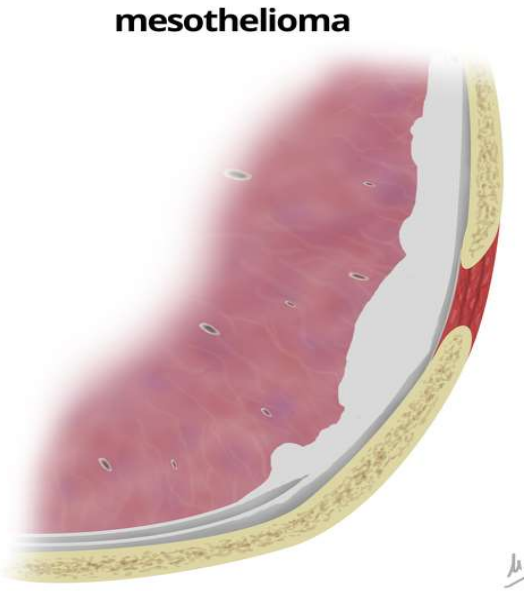
Pleural fibroma

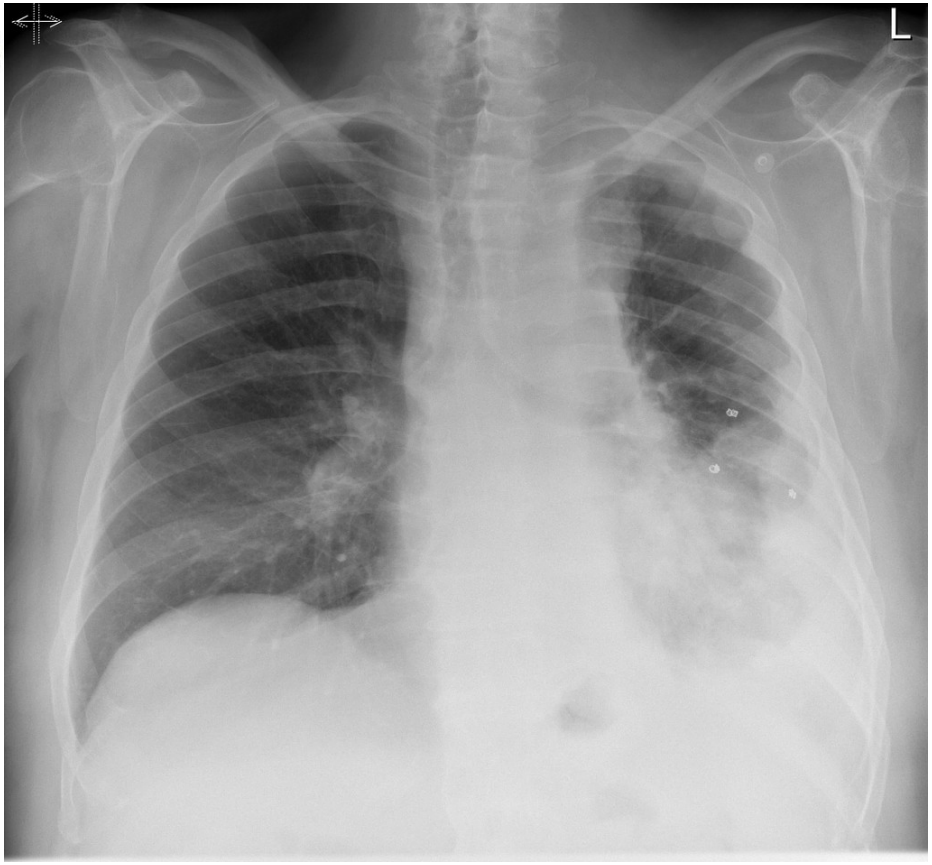


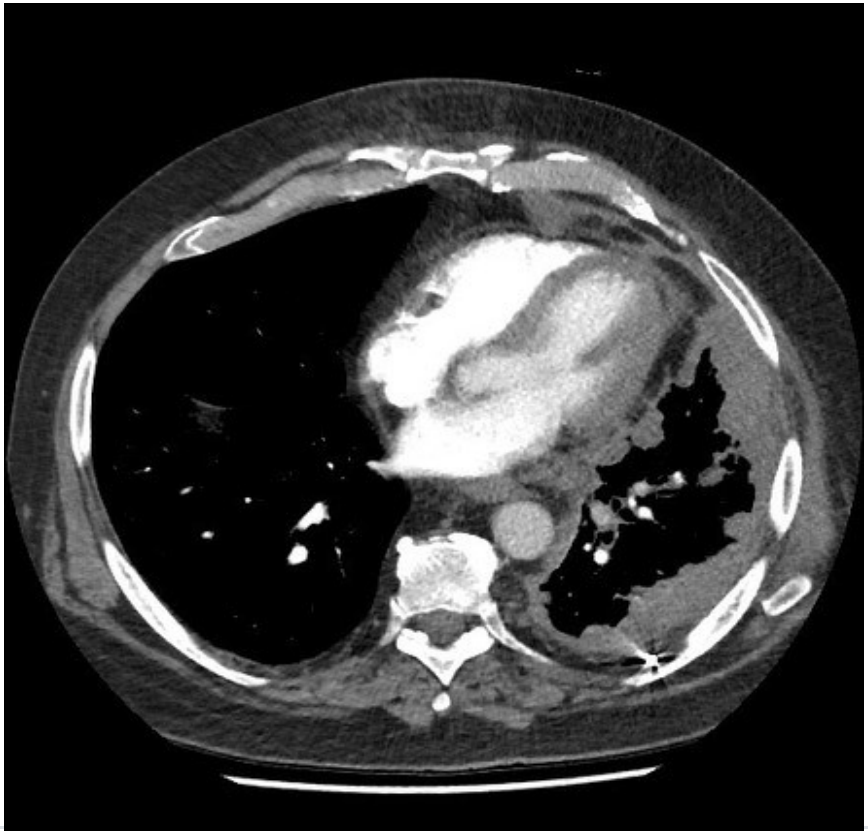
Malignant neoplasms

Pleural mesothelioma

- Is uncommon malignant tumor of the pleura.
- Strongly associated with asbestos exposure.
- It appears as pleural mass or nodular thickening of the pleura.
- Eventually grows toward lung encasement.




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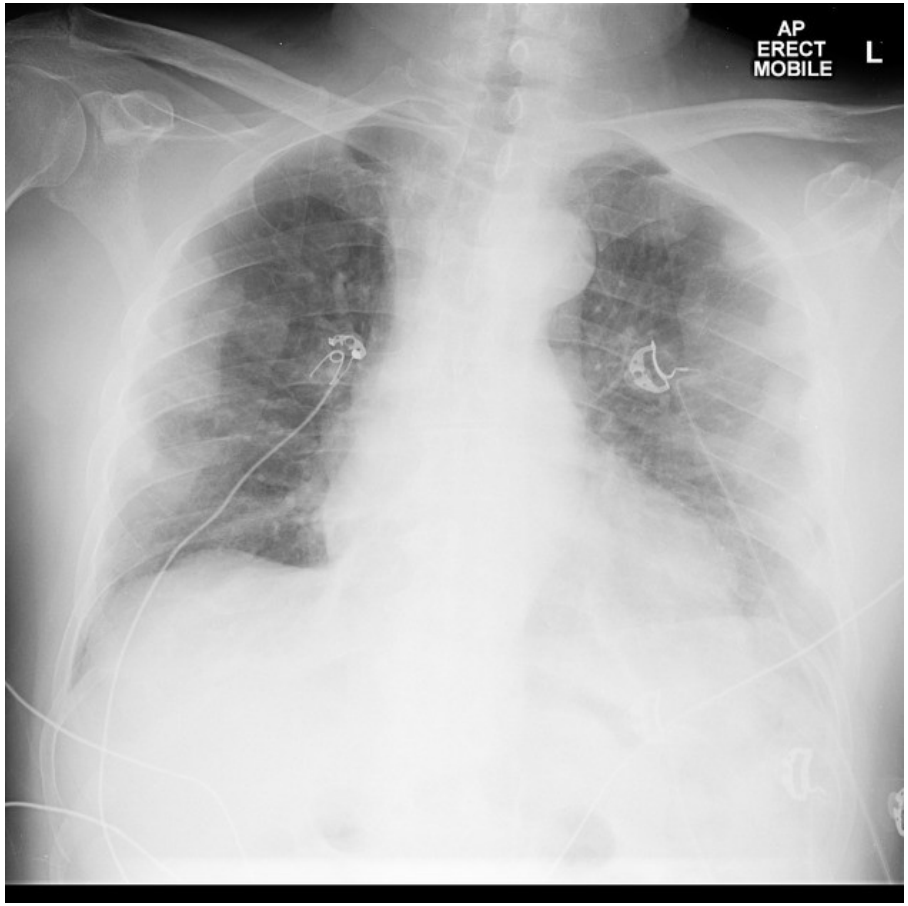


Pleural metastasis

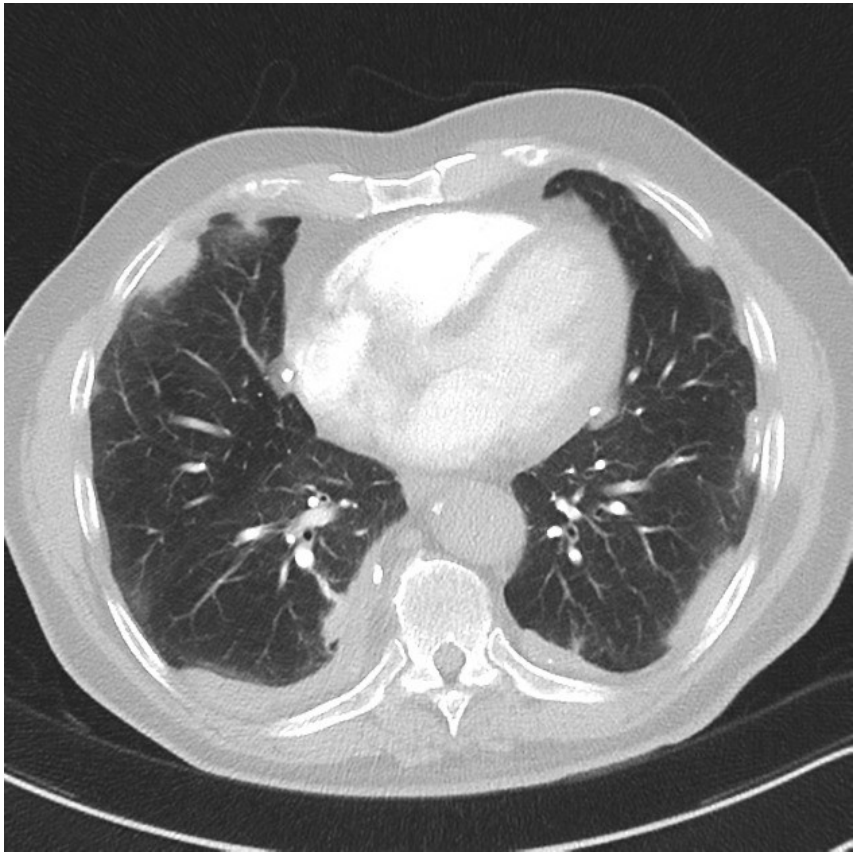
- It represents the vast majority of malignant pleural neoplasms.
- Its more common than mesothelioma.
- Usually manifested as pleural effusion.
- Lung cancer may account for up to 40% of pleural metastases and breast cancer may represents 20% of pleural metastasis.

Radiographic features:

- Pleural thickening
- Pleural masses.
- Pleural effusion.



Pleural metastasis from prostate cancer



References

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QUESTIONS & DISCUSSION

Thank you

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